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Bare laser-synthesized Si nanoparticles as functional elements for chitosan nanofiber-based tissue engineering platforms

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ABSTRACT

Methods of femtosecond laser ablation were used to fabricate bare (ligand-free) silicon (Si) nanoparticles in deionized water. The nanoparticles were round in shape, crystalline, free of any impurities, and water-dissolvable, while the dissolution rate depended on the concentration of oxygen defects in their composition. The nanoparticles were then electrospun with chitosan to form nanoparticle decorated nanofibrous matrices. We found that the functionalization of nanofibers by the nanoparticles can affect the morphology and physico-chemical characteristics of resulting nanostructures. In particular, the presence of Si nanoparticles led to the reduction of fibers thickness, suggesting a potential improvement of fiber's surface reactivity. We also observed the improvement of thermal stability of hybrid nanofibers. We believe that the incorporated Si nanoparticles can serve as functional elements to improve characteristics of chitosan-based matrices for cellular growth, as well as to enable novel imaging or therapeutic functionalities for tissue engineering applications.

Keywords: Si nanoparticles, femtosecond laser ablation and fragmentation in water, tissue engineering, chitosan, electrospinning

1. INTRODUCTION

Due to particular physico-chemical, optical, photochemical, photothermal properties, silicon (Si) nanoparticles (NPs) can be used in a range of biomedical applications, e.g., as contrast agents in bioimaging¹⁻³ based on strong room temperature photoluminescence emission⁴, photosensitizers of singlet oxygen for photodynamic therapy⁵⁻⁷, sensitizers of cancer hyperthermia under illumination by infrared radiation⁸, radio frequency radiation⁹ or ultrasound¹⁰. Besides, Si nanostructures have excellent biocompatibility and are bio-degradable¹¹, which further favors their use in biological systems. However, the synthesis of pure, uncontaminated NPs presents a real challenge as chemical¹² and electrochemical^{4-6,11} fabrication pathways can cause contamination of NPs surface by toxic by-products resulting in toxicity effects¹³. To solve the contamination problem, methods of laser ablation in gaseous¹⁴⁻¹⁸ and aqueous¹⁹⁻²⁹ media have been recently introduced. As an example, the second approach, termed as pulsed laser ablation in liquids (PLAL), implies ablation of solid target, immersed into deionized water or other "clean" liquid, which yields natural formation of NPs exempt of any hazardous contamination. Here, the employment of ultra-short laser pulses can lead to the fabrication of extremely stable colloidal solutions of low-size-dispersed crystalline NPs even in the absence of any protective ligands²³⁻²⁹. In the case of Si, such procedure can be modified by the ablation (fragmentation) from microcolloids,

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prepared preliminarily by mechanical milling of a Si wafer^{26,27,29}. Our recent data on the interaction of Si NPs with biological systems showed their negligible toxicity^{27,29}. Moreover, Si-NPs much faster biodegraded in the organism compared to porous silicon (3-5 days²⁹ compared to 4-6 weeks¹¹). Finally, laser-synthesized nanostructures provided much stronger hyperthermia effect under RF irradiation compared to electrochemically-synthesized porous silicon-based NPs⁹ and could additionally offer strong photoluminescence option in the field of relative tissue transparency³.

In our opinion, bare laser-synthesized Si NPs can also present a significant interest for tissue engineering applications. Indeed, many tissue-engineering platforms imply the use of polymeric matrices as scaffolds for the formation and growth of cellular cultures. In this case, the inclusion of functional Si NPs is expected to improve mechanical and thermal properties of the polymeric matrix, as well as enable new functionalities (e.g., imaging option³ or therapeutic effect under RF irradiation⁹). Here, chitosan is so far one of the most exploited biologically-derived polymers for these applications³⁰⁻³² due to its biocompatibility, biodegradability and bioactivity (anti-microbial, scar reduction, wound healing, etc.). Besides, the presence of amino groups on its surface enables interactions with a variety of biological species (e.g. lipids, proteins, DNA, etc.), making it a promising host structure. Chitosan-based nanofibrous matrixes, obtained by electrospinning technique, having controllable nanofiber size and porosity³³, are actively used in regenerative medicine and other tasks^{34,35}.

In our recent study³⁶, we reported the functionalization of chitosan nanofibers by bare laser-synthesized Au and Si NPs and testing of such hybrid structures as potential matrices for tissue engineering tasks. In this work, we present details of characterization of laser-synthesized Si NPs alone or in combination with chitosan polymers (TEM, ζ – potential, confocal Raman, SEM, TGA and DSC etc.), as well as their exploratory testing as functional material for chitosan based nanofibrous matrices.

2. EXPERIMENTAL METHODS

Si NPs were prepared by pulsed laser ablation in water, using radiation from a Yb:KGW laser (Amplitude Systems, 1025 nm, 480 fs, 1 kHz). Initial Si micropowder was preliminary prepared by mechanical milling of a Si wafer.^{26,27} 1.4 mg of powder was then dispersed in a glass cuvette, filled with 4 ml of deionized water (18 M Ω ·cm at 25°C), using an ultrasonication bath for 30 minutes. Then, obtained Si microcolloid was fragmented under laser irradiation for one hour. The laser beam was focused, using a lens 75 mm focusing distance at 1 cm below the water level. The solution was continually homogenized by a magnetic stirrer during fragmentation process. The experiments were carried out at ambient conditions (oxygen-rich) and after bubbling solution with Argon (Ar) gas (oxygen-free), similarly to how it was done in Ref. 27. The formed Si colloids were then transferred into centrifugal plastic tubes (Cellestar®tubes) and centrifuged at 3500 rpm for 20 minutes to remove possible unfragmented Si micropowder. The concentration of the Si NPs in the solution was measured by inductively coupled plasma mass spectroscopy (ICP-MS) method.

Climate-controlled electrospinning platform from IME technologies was employed to elaborate the nanofibers. Commercial chitosan powder (Mw = 200 kD) with 75% of deacetylation was used (Sigma-Aldrich). Poly(ethylene oxide) (PEO) powder (Mw = 300 kD) and acetic acid (ReagentPlus®, 99%) solution were purchased respectively from Sigma Aldrich and Normapur (PROLABO). Electrospinning process was performed using a horizontal setup. Briefly, the prepared solution was transferred into a 3 ml syringe (IME technologies) with Luer-Lock connected via a PTFE tube to flat needles. The syringe was mounted on a programmable pumps module to ensure a steady electrospinning flow and connected to a rotating target collector. The flow rate was fixed at 0.3ml·h⁻¹ for all experiments. Nanofibers were collected on an aluminum foil placed around a rotating target collector equipped with nozzle stage. The speed of the rotating collector and the nozzle translation were 80 rpm and 10 mm s⁻¹, respectively. Voltages applied to the solution and the collector was fixed at 12 kV and -1 kV, respectively. All experiments were conducted under normal atmospheric conditions at 28 °C temperature and a relative humidity of 30%.

A high-resolution transmission electron microscopy (HR-TEM) system (JEOL JEM 3010) was employed in imaging and diffraction modes in order to study size dispersion of Si NPs and their crystalline structure. A droplet of Si NPs colloid was drop casted onto the carbon-coated TEM copper grid and then dried at ambient conditions. ζ – potential of NPs in water at neutral pH was measured using a Zetasizer (Nano ZS, Malvern, UK).

Raman measurements were performed using NT-MDT Ntegra Spectra C confocal Raman system. An inverted optical scheme was used for sample analysis (both excitation and emission signal were collected from the substrate side of the sample). An oil immersion objective (Olympus, 100x, 1.3 NA) was used for focusing 532 nm CW laser radiation on the sample. Laser radiation power was set at 20 μ W to avoid heat-related effects during the measurements. Samples were

prepared by drop casting Si NPs colloid on 150 μm thick borosilicate glass substrate and their subsequent drying under ambient conditions.

Dissolution kinetics of Si NPs was studied via dialysis experiments. Dialysis sack (12–14 kD, Serva) was filled with 10 mL of Si NPs colloid and transferred into glass vessel with 4 L of deionized water (18 $\text{M}\Omega\cdot\text{cm}$ at 25°C). Continuous stirring of water using magnetic stirrer was employed to ensure a constant concentration gradient at walls of dialysis sack during the dissolution process. The suspensions of Si NPs were examined by TEM several times in the course of dissolution experiment.

The size and the surface morphologies of the nanofibers matrixes were characterized by a scanning electron microscope (SEM), using a DSM 982 Gemini (Zeiss, Germany) microscope at accelerating voltage of 15 kV. Samples were fixed on an aluminum foil to reduce charging of polymeric fibers. High magnification measurements were done using a Joel JSM-6320F microscope operating at 0.5–30 kV, coupled with energy dispersive X-ray analysis (EDX) to detect Si NPs.

Thermal decomposition profiles of the blend nanofibers were characterized by TGA analysis performed on Perkin-Elmer system (TGA 4000). For each composition, 12 mg of Si NPs modified nanofibers were taken and placed into the TGA with a nitrogen atmosphere at a flow rate of 20 ml min^{-1} and at heating rate of 10 $^{\circ}\text{C min}^{-1}$ in a temperature range from 25 $^{\circ}\text{C}$ to 800 $^{\circ}\text{C}$. Differential scanning calorimetry (DSC) was also performed on the elaborated nanofibers. The specimens were equilibrated to 30 $^{\circ}\text{C}$ and then heated up to 250 $^{\circ}\text{C}$. The measurements were repeated three times.

3. RESULTS AND DISCUSSION

Before mixing with chitosan/PEO solution and electrospinning, bare NPs were characterized by TEM and Raman. As shown in Fig. 1, NPs were typically spherical with mean size 33 nm, size dispersion 40 nm full width at half maximum (FWHM). Crystallinity was examined by electron diffraction and confocal Raman (Fig. 1b). Diffraction data clearly demonstrated several diffraction rings, corresponded to different Si crystallographic planes, while Raman scattering spectrum had sharp peak, centered at 520 cm^{-1} with slight widening toward lower energy, which suggested polycrystallinity of Si NPs with certain amorphous phase fraction. Thus, measurements which revealed predominantly crystalline structure of NPs. ζ – potential measurements were carried out in water-based colloid at natural pH, revealing negative surface charge of NPs (-45 ± 1.5 mV) due to partial oxidation of their surface. We earlier showed in Ref. 26 that Si NPs prepared by laser fragmentation from microcolloids are free of any contamination.

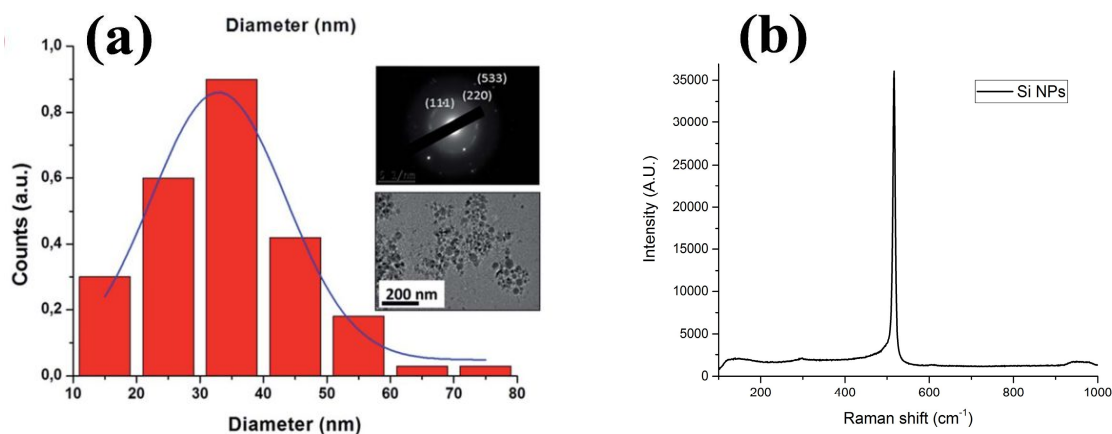


Figure 1. (a) Typical TEM and EDX images of Si NPs (insets) and corresponding size distribution. (b) Raman spectrum of Si NPs.

The biodegradability of nanoparticles is one key parameters related to their biomedical applications. We checked water solubility of Si NPs, obtained under oxygen-rich and oxygen-free conditions. Briefly, setup for dialysis experiment (as described in experimental methods section) was used for monitoring nanoparticles size for 30 days²⁷. Results are presented in Fig. 2. Si NPs prepared at oxygen-rich condition dissolved much faster due to high amount of oxygen-related defects in their core. Thus, control of Si NPs dissolution rate is available through changing the concentration of dissolved oxygen. Three formulations of chitosan(PEO)/NPs nanofibers were prepared with different concentrations of Si NPs. Nomenclature and compositions used are summarized in Table 1. Nanofibers were prepared from optimized chitosan/PEO ratio of 0.25. Polymers were then dissolved in 10 ml of 90% acetic acid solution. Electrospinning of pure

chitosan dissolved in the range of 0.5–2 wt% was unsuccessful, resulting in the formation of nanobeads instead of nanofibers. Several studies reported the difficulty to electrospin pure chitosan solution, mainly because of its polycationic nature and rigid chemical structure due to the formation of specific inter and intra-molecular interactions (strong hydrogen bonds) preventing free movements of polymeric chain segments leading to explosion-like behavior of the jet.^{37,38} On the other hand, chitosan nanofibers were obtained by incorporating poly(ethylene oxide) PEO as co-spinning agent. A continuous jet was thus observed leading to the formation of uniform nanofibers on the collector. PEO can act as a plasticizer by breaking down inter and intra molecular interactions of chitosan chains.^{39,40} The interaction of PEO with chitosan chains through hydrogen bonds ensure the formation of chitosan(PEO) – based nanofibrous matrix.

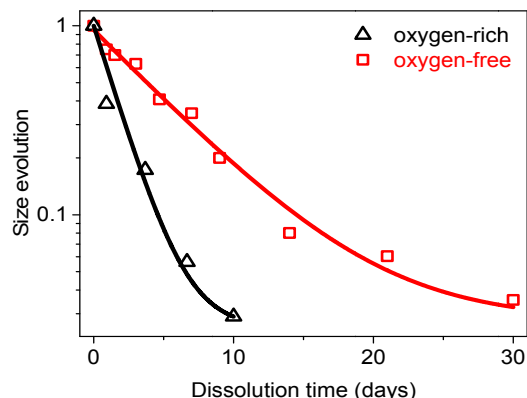


Figure 2. Time-dependent size evolution of Si NPs, obtained under oxygen-rich and oxygen-free conditions.

Table 1. Polymer composition of hybrid chitosan(PEO) nanofibers prepared with different concentrations of Si NPs.

Sample nomenclature	Reference	Polymer composition wt (%)		
		Chitosan	PEO	Si NPs
Chitosan(PEO)	Ch-0	25	75	-
Chitosan(PEO)-SiNPs	Ch-10Si	22.5	67.5	10
Chitosan(PEO)-SiNPs	Ch-30Si	17.5	52.5	30

SEM observations showed that the nanofibers exhibit cylindrical morphology with mean diameter of 687 ± 120 nm, forming a dense network with no trace of beads (Fig. 3a). The functionalization of chitosan(PEO) fibers with Si NPs was done at 10 and 30 wt%, while all the electrospinning parameters were constant. The addition of NPs had no effect on the chitosan(PEO) nanofiber morphology, while the presence of spherical NPs on the surface of the nanofibers was clearly visible. However, functionalization by Si NPs led to almost two times reduction of nanofibers mean size (Fig. 3b) from 687 ± 120 nm to 386 ± 80 nm. These observations may be attributed to electrostatic interactions between positive group (NH_3^+) of the nanofiber's surface and the negative surface charge of NPs (SiO_x^-).^{36,37}

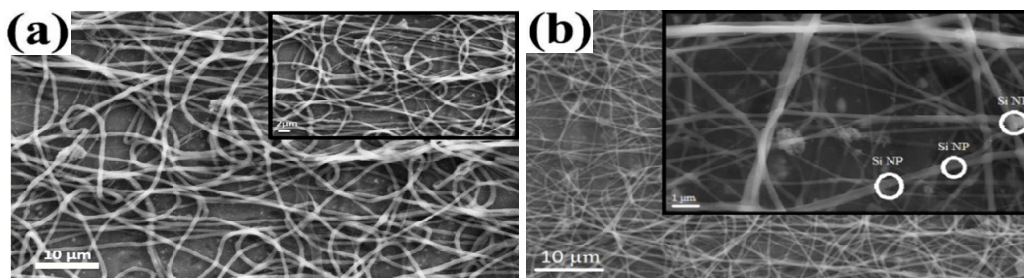


Figure 3. SEM micrographs of (a) pure chitosan(PEO) fibers and (b) hybrid chitosan(PEO) nanofibers functionalized with Si NPs

TGA characterization of nanofibers, conjugated with Si NPs is presented in Fig. 4. The measurements were performed at heating rate of $10\text{ }^{\circ}\text{C min}^{-1}$ in $25\text{--}800\text{ }^{\circ}\text{C}$ interval. Chitosan(PEO) (Ch-0) displayed one weight loss region at $383\text{ }^{\circ}\text{C}$ corresponding to the loss of 95.44% of initial mass and attributed to the degradation of the polymer molecules.³⁹

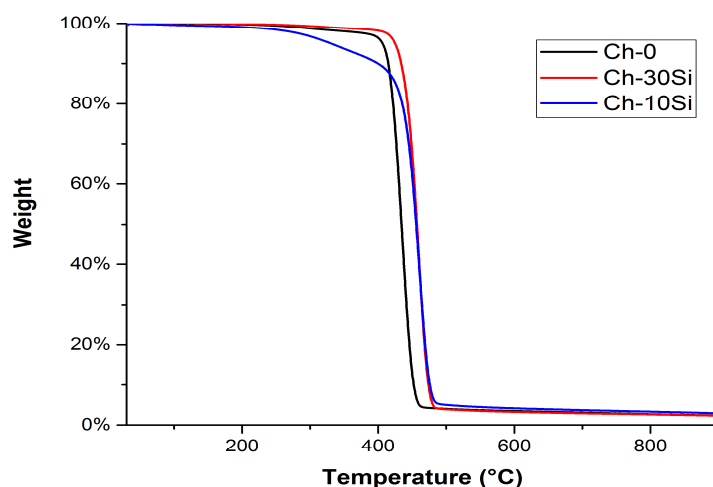


Figure 4. TGA thermogram curves of hybrid chitosan(PEO) nanofibers functionalized by Si NPs.

The functionalization of the nanofibers with Si NPs, led to a shift of the degradation temperature to higher values, although a slow weight loss too place at lower temperatures. A fast decomposition event was observed at $417\text{ }^{\circ}\text{C}$ and $425\text{ }^{\circ}\text{C}$, respectively, for Ch-10Si and Ch-30Si. This slight shift of degradation temperature suggests a better thermal stability, which is probably due to higher heat capacity of Si NPs. Based on this hypothesis, it will be interesting to elaborate hybrid chitosan(PEO) with higher content of NPs to highlight this aspect.

Thermal gravimetric analysis was also performed by differential scanning calorimetry (DSC). These results are presented in Fig. 5. A classical DSC curve was observed with two events for Ch-0 nanofibers. The first one occurred at $56.4\text{ }^{\circ}\text{C}$, corresponding to latent heat of melting, attributed in the literature to favorable interaction between crystalline PEO and amorphous chitosan chain to form strong intermolecular hydrogen bonds between the ether groups of PEO and the amino/hydroxyl groups of chitosan.^{41,42} At higher temperature around $125\text{ }^{\circ}\text{C}$, a second predominant event related to a degradation of the polymer matrix takes place.

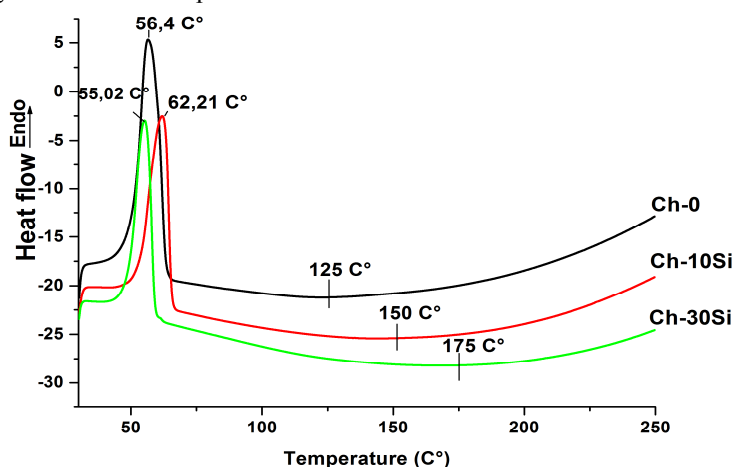


Figure 5. DSC thermogram curves of hybrid chitosan(PEO) nanofibers functionalized with Si NPs

The addition of Si NPs (Ch-10Si and Ch-30Si) into the initial formulation had no significant influence on the first endothermic peak. Here, the first peak is slightly shifted to $62.21\text{ }^{\circ}\text{C}$ and $55.02\text{ }^{\circ}\text{C}$ for Ch-10Si and Ch-30Si respectively,

compared to 56.4 °C for Ch-0. However, the intensity of the peaks decreased when chitosan nanofibers were functionalized with Si NPs, which could be explained by a lower mass loss resulting in a better thermal stability. Such an assumption is supported by the increase of endothermic enthalpy value ranging from 185.81 Jg⁻¹ for Ch-0 to 364.87 Jg⁻¹ for Ch-10Si. We suppose that the same increase of enthalpy is responsible for the shift of the second endothermic peak to higher temperatures.

4. CONCLUSIONS

We demonstrated the possibility of functionalization of chitosan nanofibers with bare Si NPs prepared by pulsed laser ablation in liquid. Detailed analysis showed that spherical Si NPs with a crystalline structure are coating cylindrical chitosan fibers via, most probably, electrostatic interaction between positively charged amine groups of chitosan and negatively charged surface of NPs. The introduction of Si NPs led to decrease of nanofibers width, which may improve their biological activity. Another important result is reinforced thermal stability of such hybrids, confirmed by TGA and DSC characterizations. These data, complemented by a well-known biocompatibility, biodegradability and rich imaging and therapeutic modalities of laser synthesized Si NPs, promises exciting perspectives for such chitosan-based hybrids in tissue engineering applications.

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